



Penelusuran/Searching: Informed Search

KOM321 Pertemuan 4 MK Kecerdasan Buatan

Informed Search

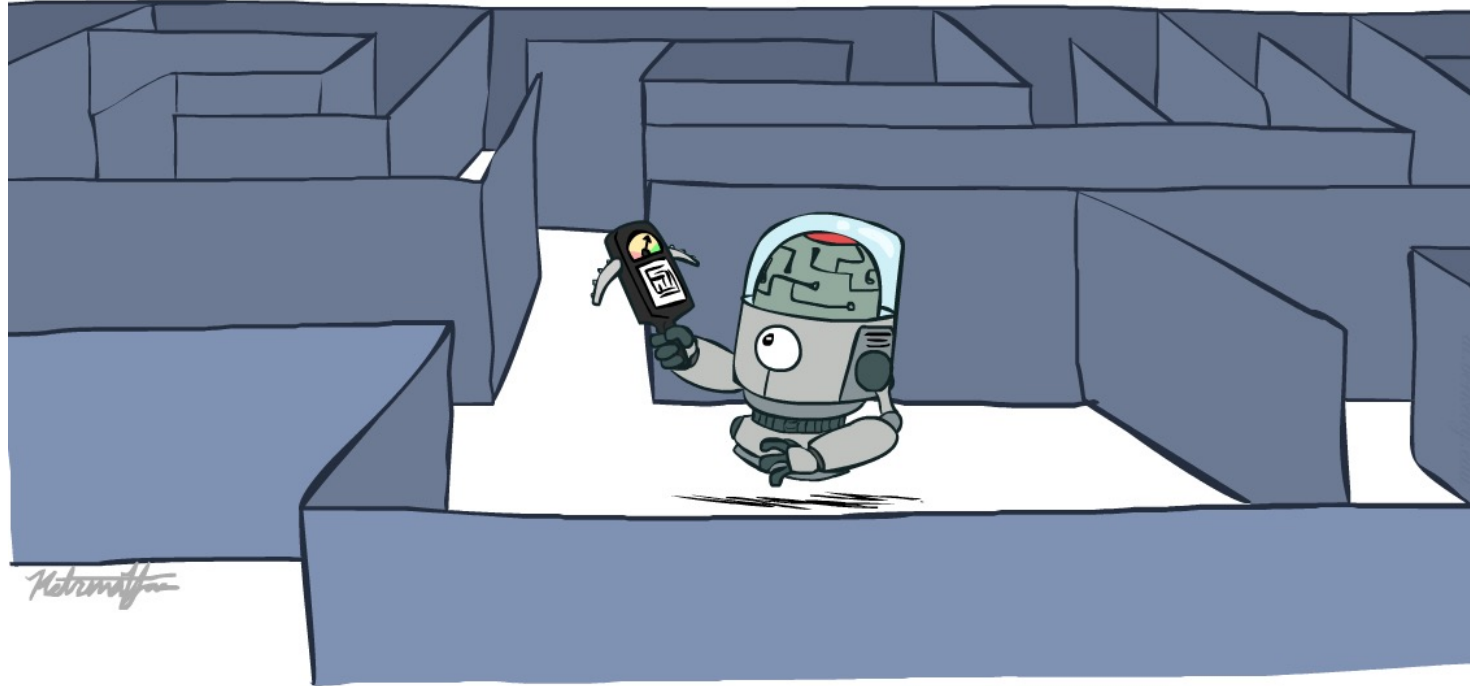


Illustration taken from slides by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley.

(<http://ai.berkeley.edu>)



Apa yang akan kita pelajari hari ini?

1. *Recap: Search*
Example: Pancake Problem
2. *Informed Search*
 - Heuristics
 - Greedy Search
 - A* Search
3. *A* Applications*



Apa yang akan kita pelajari hari ini?

1. *Recap: Search*

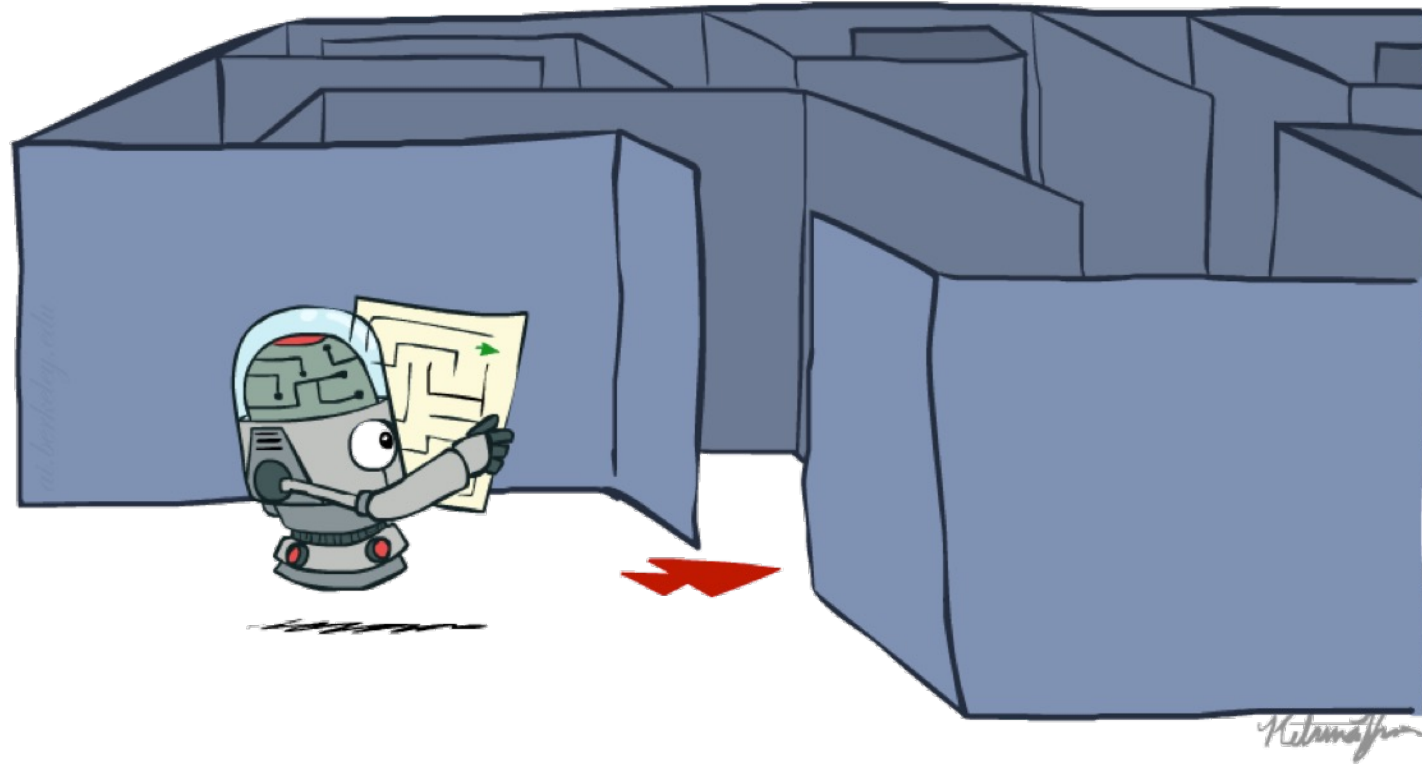
Example: Pancake Problem

2. *Informed Search*

- Heuristics
- Greedy Search
- A* Search

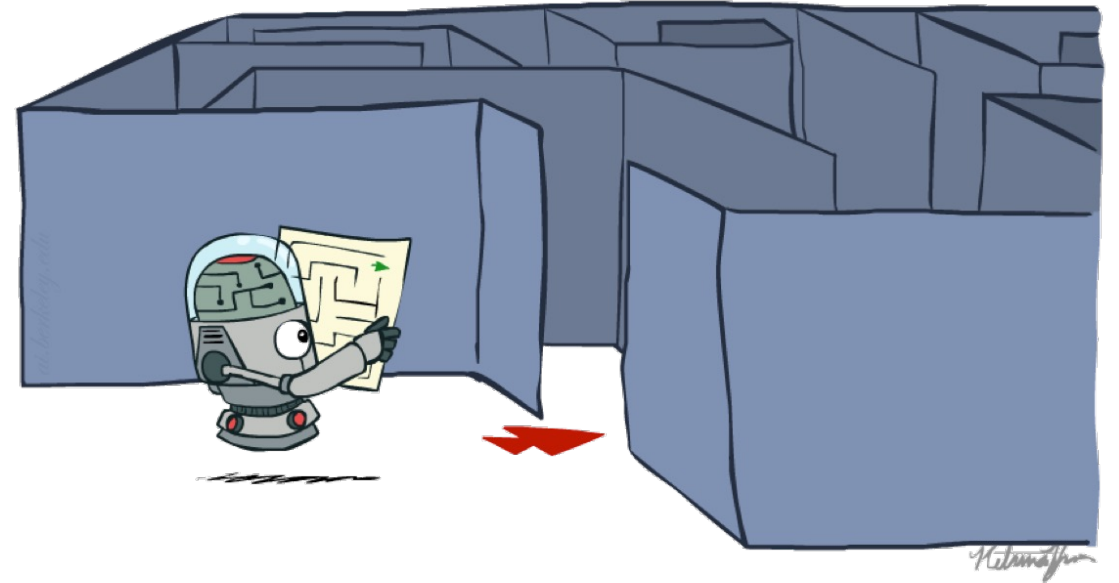
3. *A* applications*

Recap: Search



Recap: Search

- **Search problem:**
 - States (configurations of the world)
 - Actions and costs
 - Successor function (world dynamics)
 - Start state and goal test
- **Search tree:**
 - Nodes: represent plans for reaching states
 - Plans have costs (sum of action costs)
- **Search algorithm:**
 - Systematically builds a search tree
 - Chooses an ordering of the fringe (unexplored nodes)
 - Optimal: finds least-cost plans

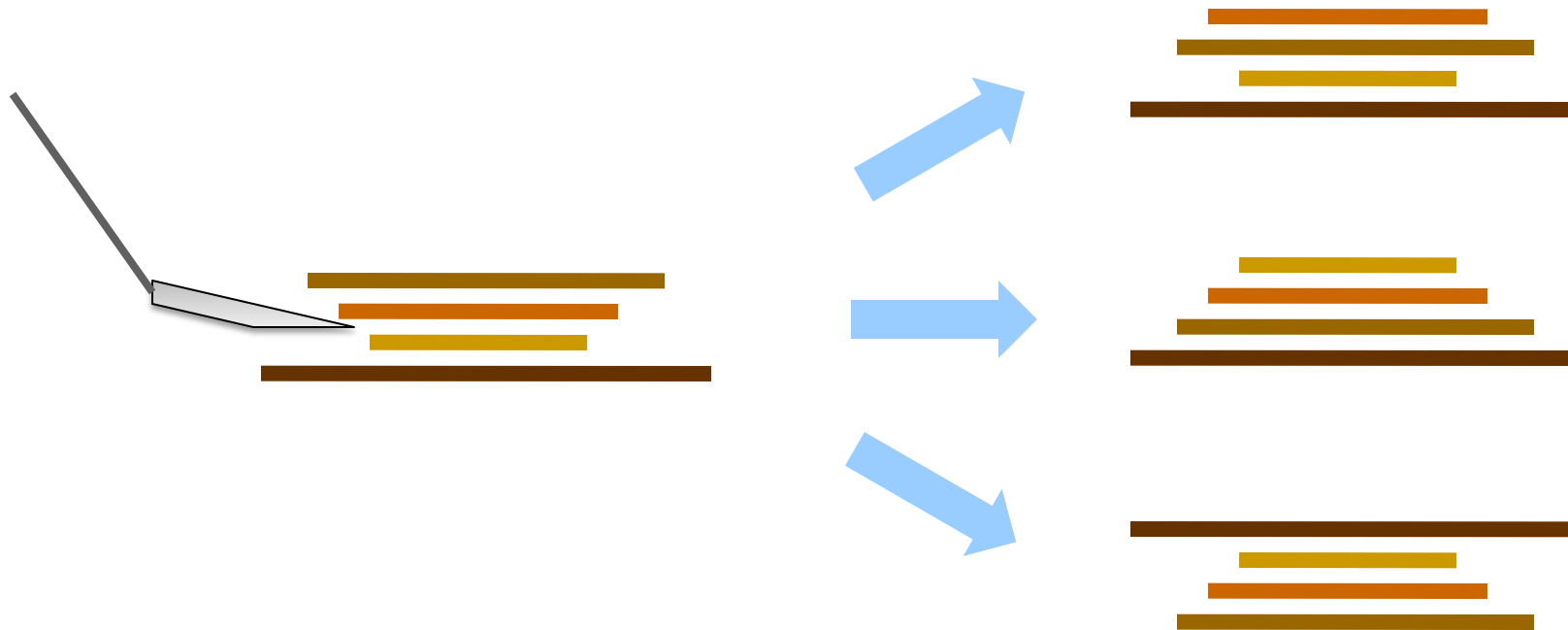




Apa yang akan kita pelajari hari ini?

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3. *Graph Search*

Example: Pancake Problem



Cost: Number of pancakes flipped

Example: Pancake Problem

BOUNDS FOR SORTING BY PREFIX REVERSAL

William H. GATES

Microsoft, Albuquerque, New Mexico

Christos H. PAPADIMITRIOU*†

Department of Electrical Engineering, University of California, Berkeley, CA 94720, U.S.A.

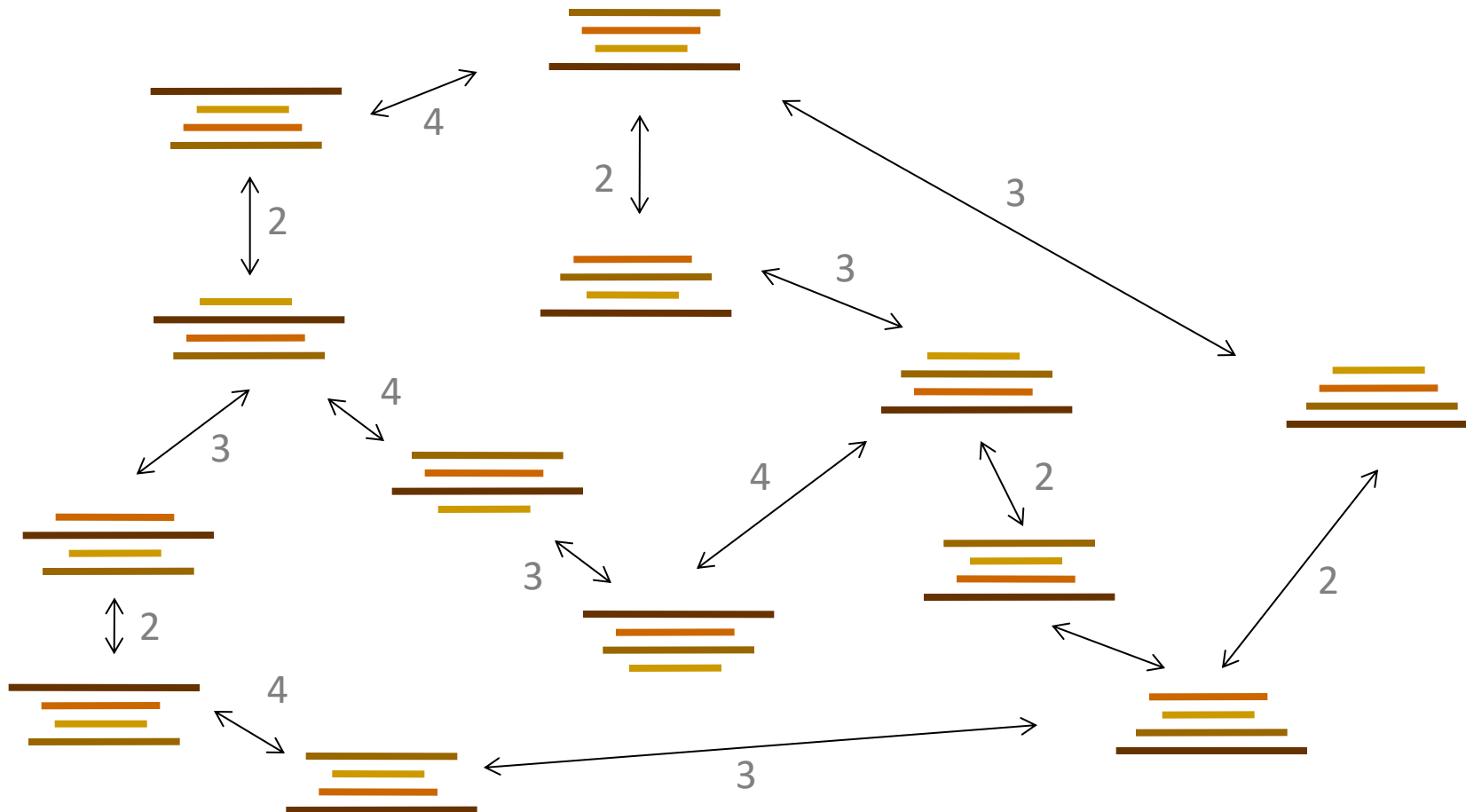
Received 18 January 1978

Revised 28 August 1978

For a permutation σ of the integers from 1 to n , let $f(\sigma)$ be the smallest number of prefix reversals that will transform σ to the identity permutation, and let $f(n)$ be the largest such $f(\sigma)$ for all σ in (the symmetric group) S_n . We show that $f(n) \leq (5n+5)/3$, and that $f(n) \geq 17n/16$ for n a multiple of 16. If, furthermore, each integer is required to participate in an even number of reversed prefixes, the corresponding function $g(n)$ is shown to obey $3n/2 - 1 \leq g(n) \leq 2n + 3$.

Example: Pancake Problem

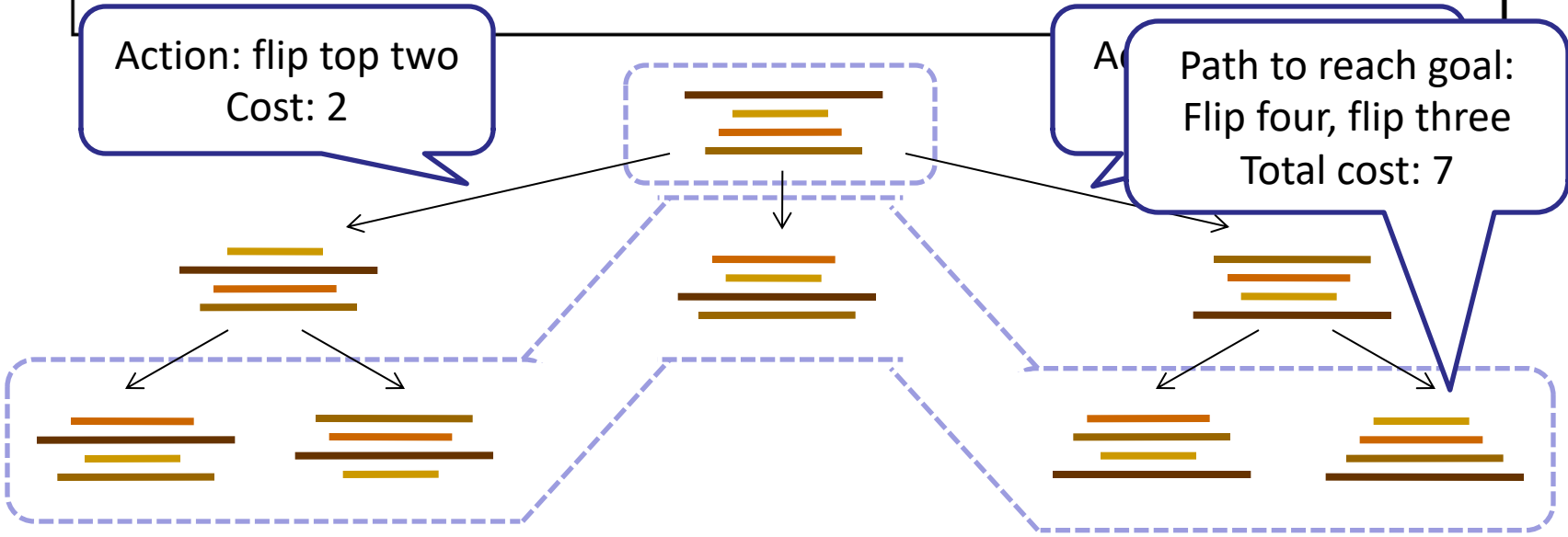
State space graph with costs as weights



```

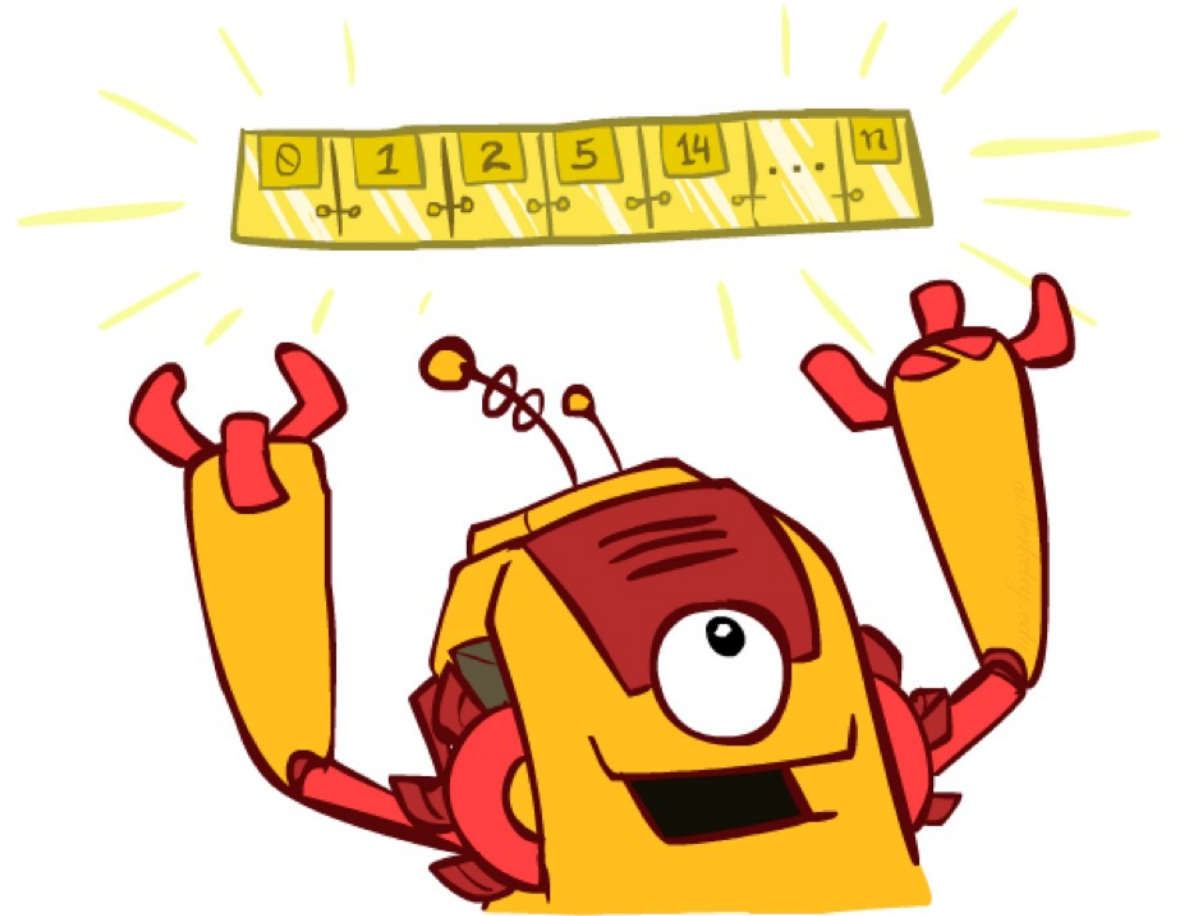
function TREE-SEARCH(problem, strategy) returns a solution, or failure
    initialize the search tree using the initial state of problem
    loop do
        if there are no candidates for expansion then return failure
        choose a leaf node for expansion according to strategy
        if the node contains a goal state then return the corresponding solution
        else expand the node and add the resulting nodes to the search tree
    end

```



The One Queue

- All these search algorithms are the same except for fringe strategies
 - Conceptually, all fringes are priority queues (i.e. collections of nodes with attached priorities)
 - Practically, for DFS and BFS, you can avoid the $\log(n)$ overhead from an actual priority queue, by using stacks and queues
 - Can even code one implementation that takes a variable queuing object

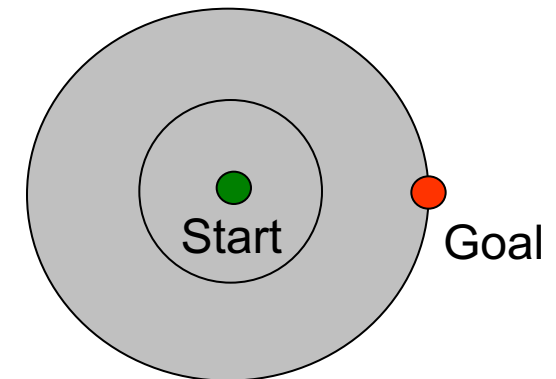
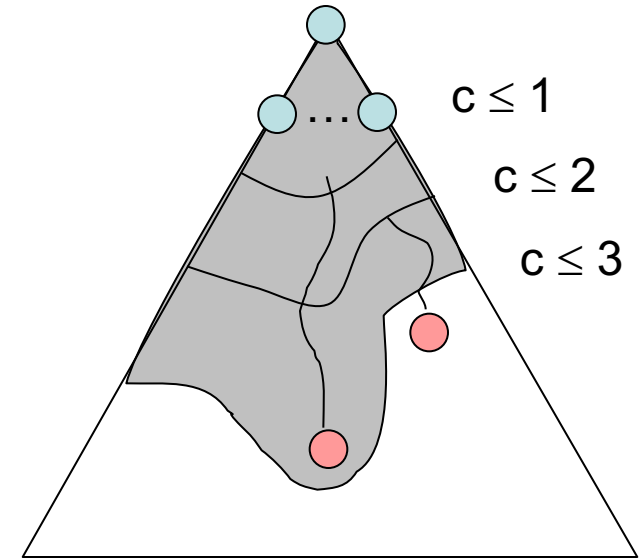


Uninformed Search



Uniform Cost Search

- Strategy: expand lowest path cost
- The good: UCS is complete and optimal!
- The bad:
 - Explores options in every “direction”
 - No information about goal location



[Demo: contours UCS empty (L3D1)]

[Demo: contours UCS pacman small maze (L3D3)]

Video of Demo Contours UCS Empty



Video of Demo Contours UCS Pacman Small Maze





Apa yang akan kita pelajari hari ini?

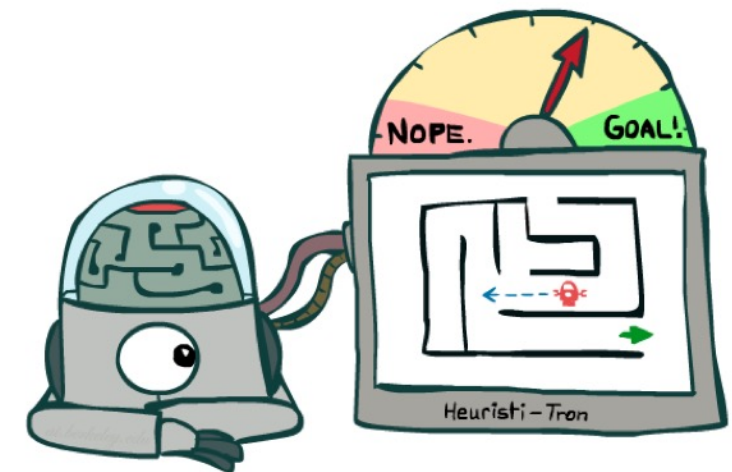
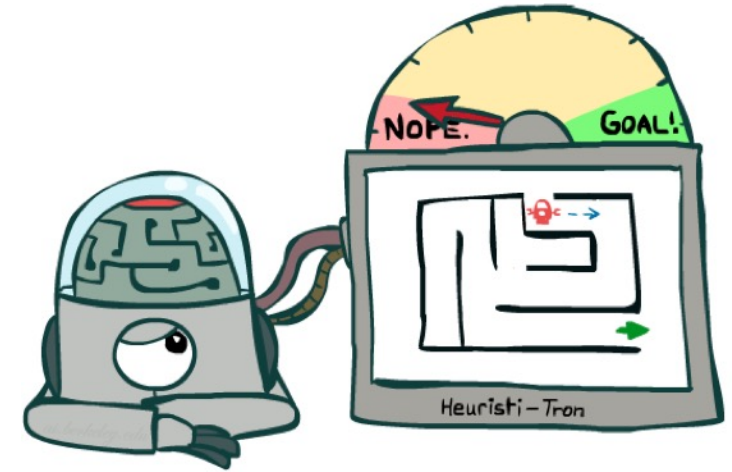
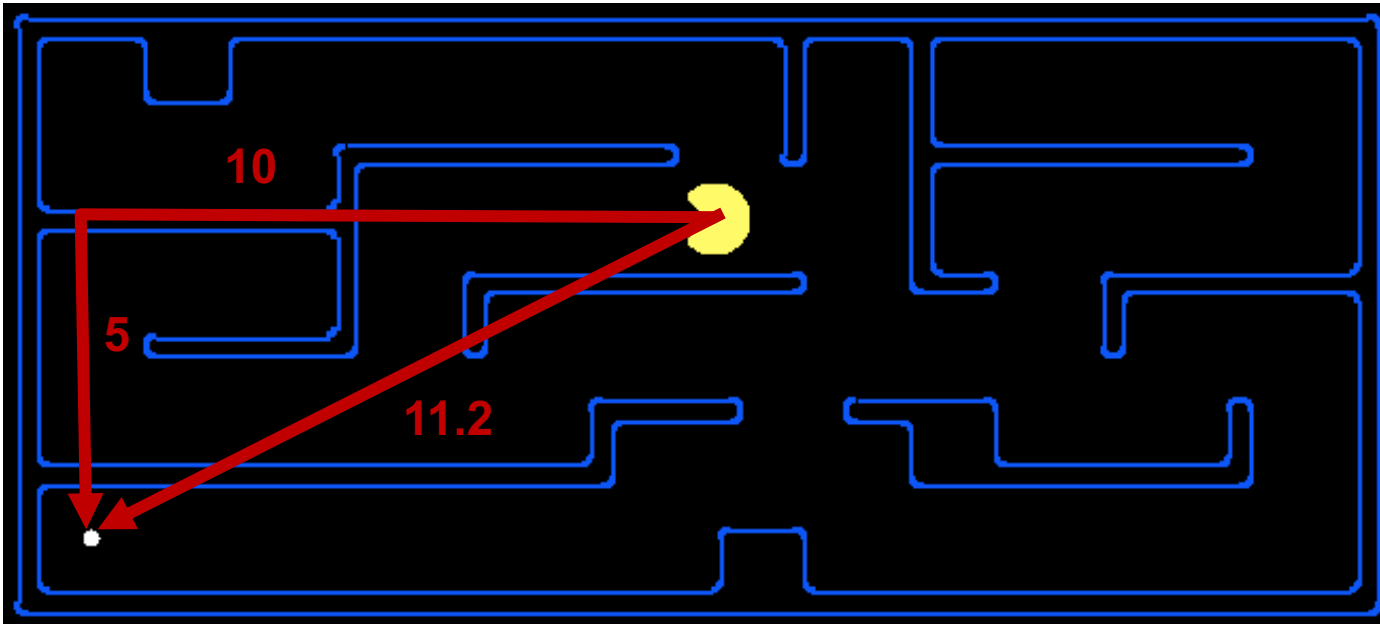
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Informed Search

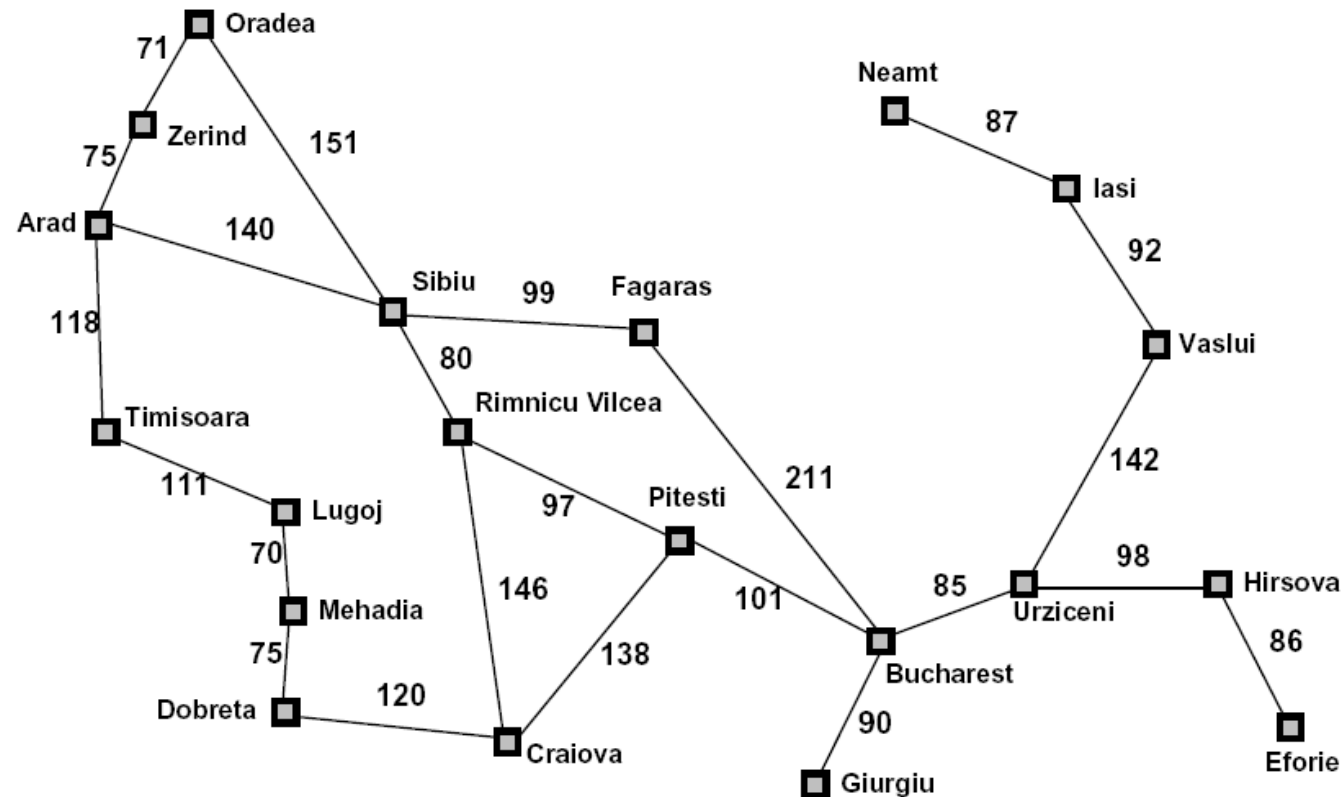


Search Heuristics

- A heuristic is:
 - A function that *estimates* how close a state is to a goal
 - Designed for a particular search problem
 - Examples: Manhattan distance, Euclidean distance for pathing



Example: Heuristic Function



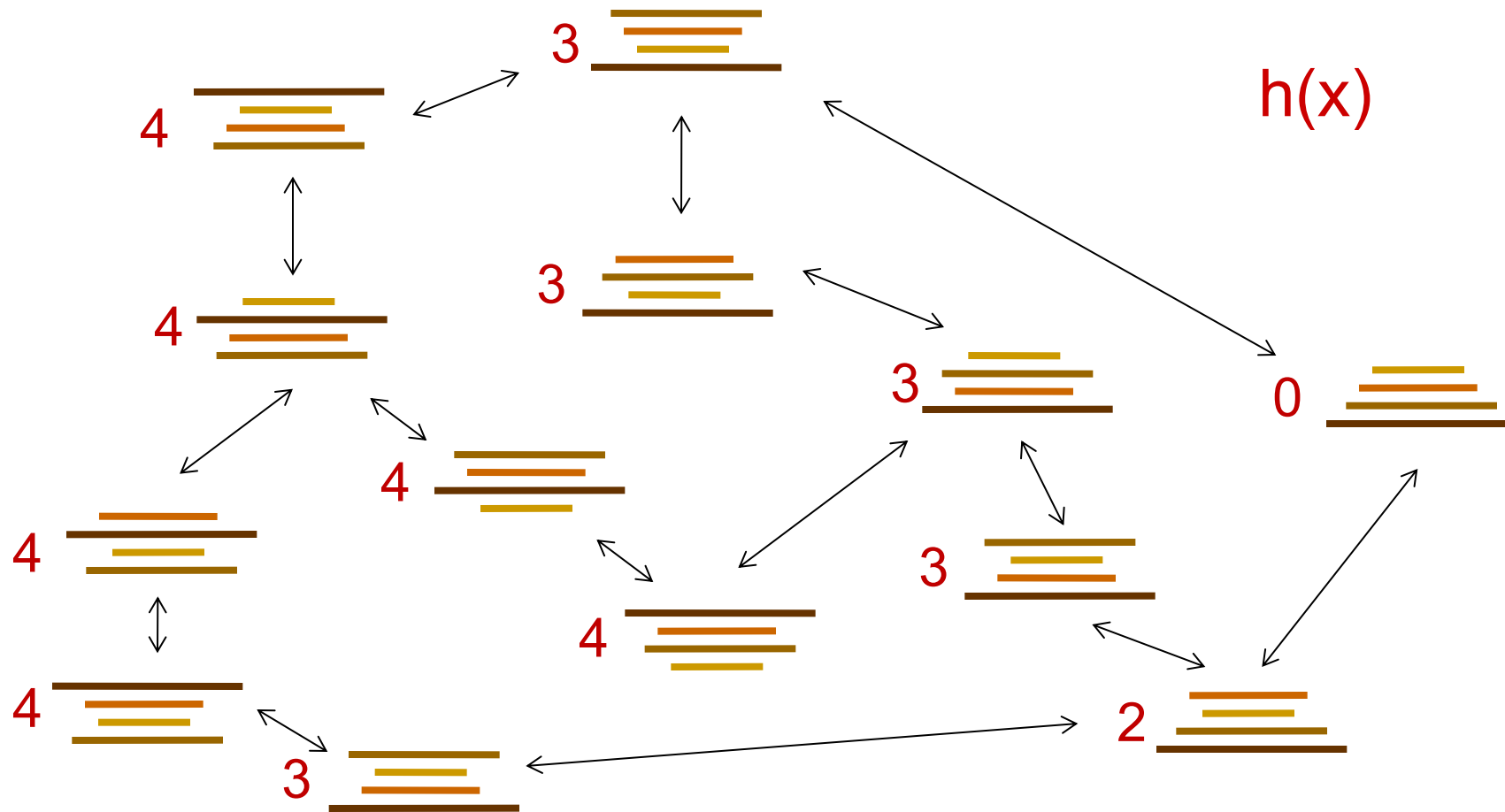
Straight-line distance
to Bucharest

Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

$h(x)$

Example: Heuristic Function

Heuristic: the number of the largest pancake that is still out of place





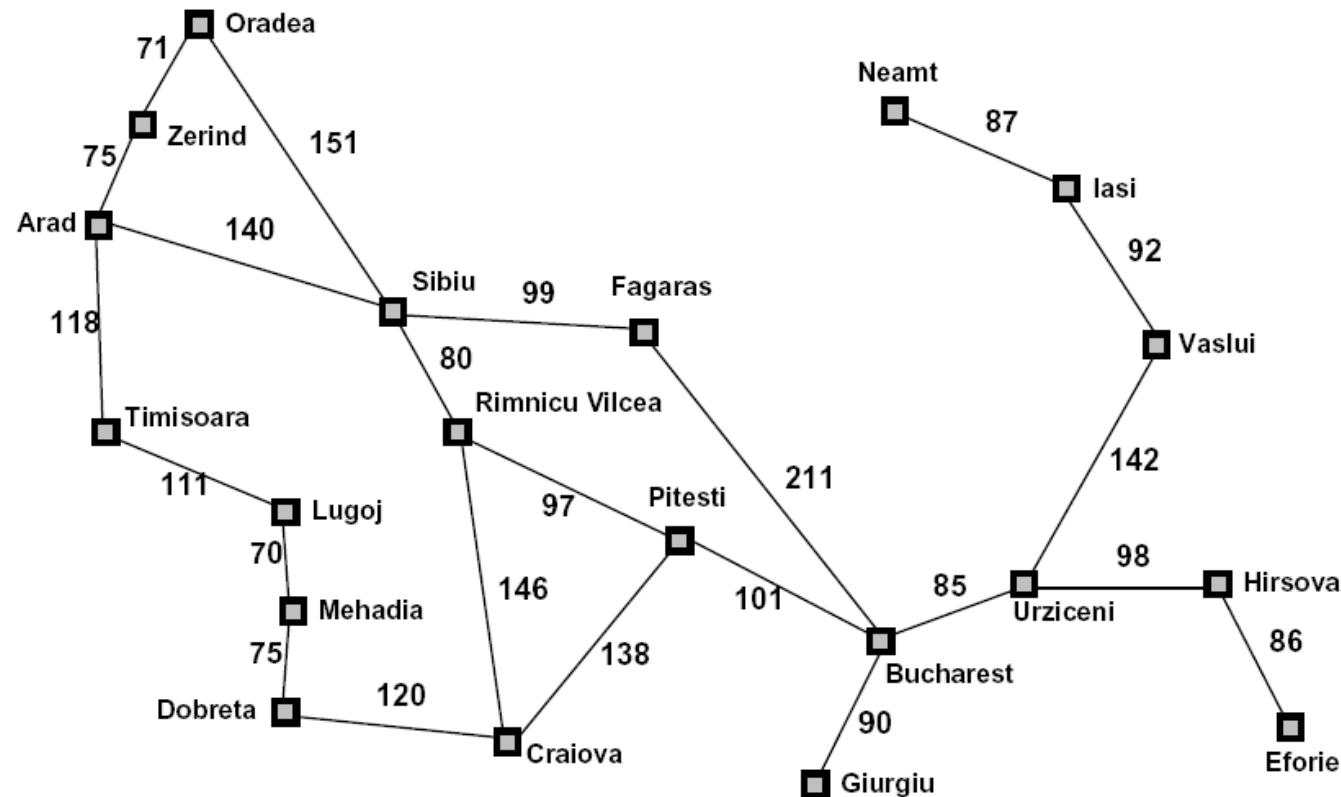
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Greedy Search



Example: Heuristic Function



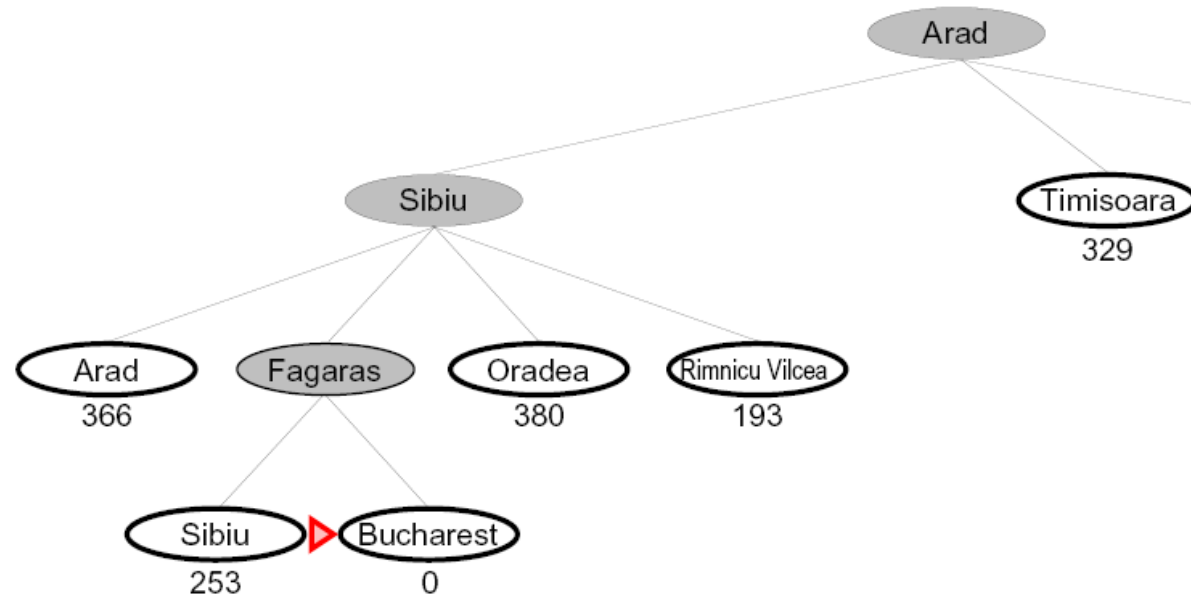
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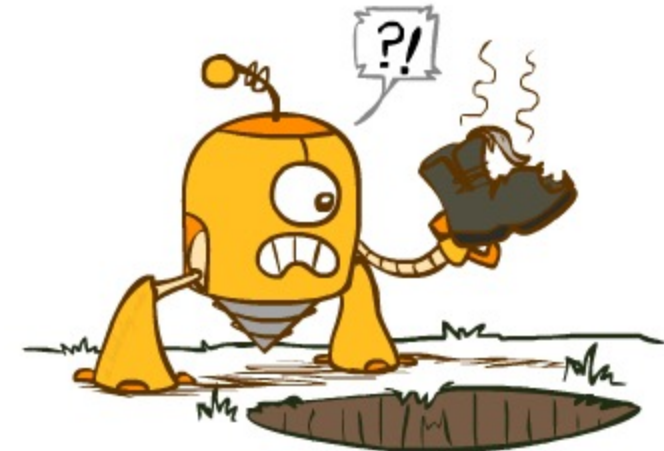
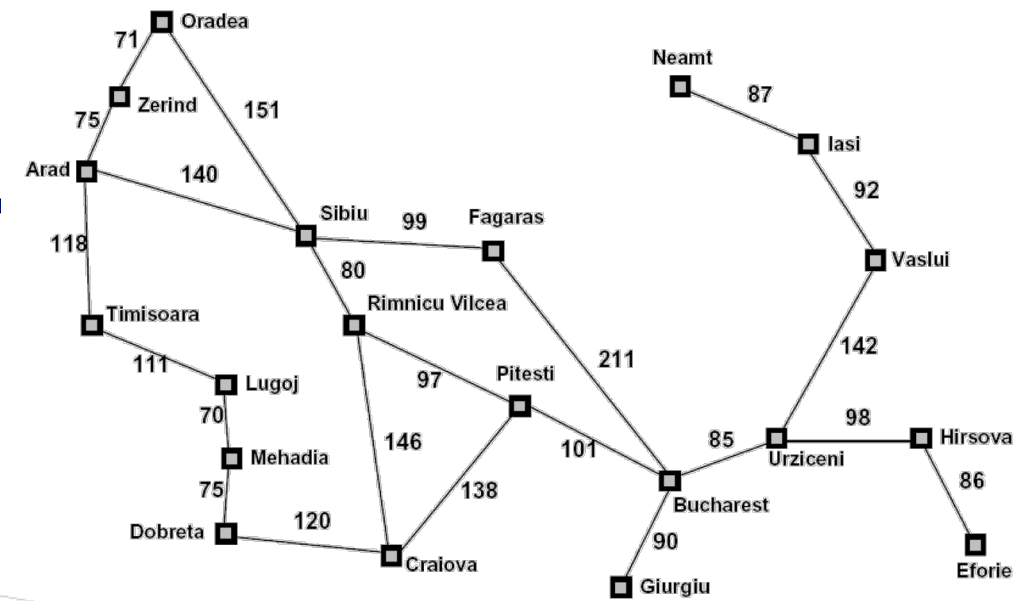
$h(x)$

Greedy Search

- Expand the node that seems closest...

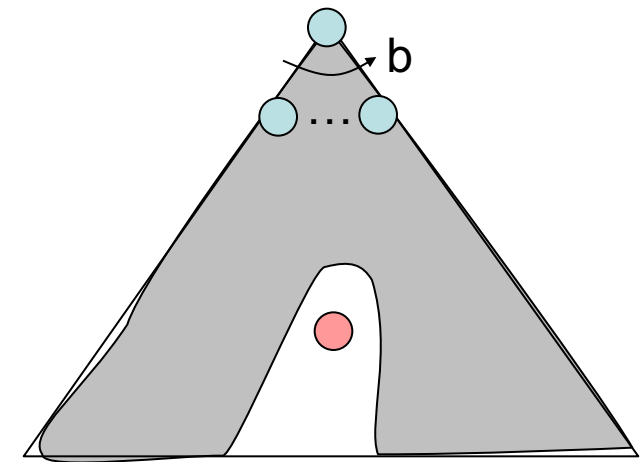
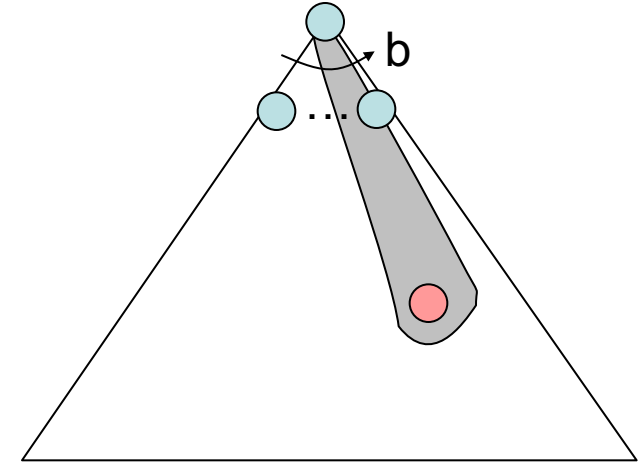


- What can go wrong?



Greedy Search

- Strategy: expand a node that you think is closest to a goal state
 - Heuristic: estimate of distance to nearest goal for each state
- A common case:
 - Best-first takes you straight to the (wrong) goal
- Worst-case: like a badly-guided DFS



[Demo: contours greedy empty (L3D1)]

[Demo: contours greedy pacman small maze (L3D4)]

Video of Demo Contours Greedy (Empty)



Video of Demo Contours Greedy (Pacman Small Maze)





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A* Search



A* Search

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2/2/20

3/3/20

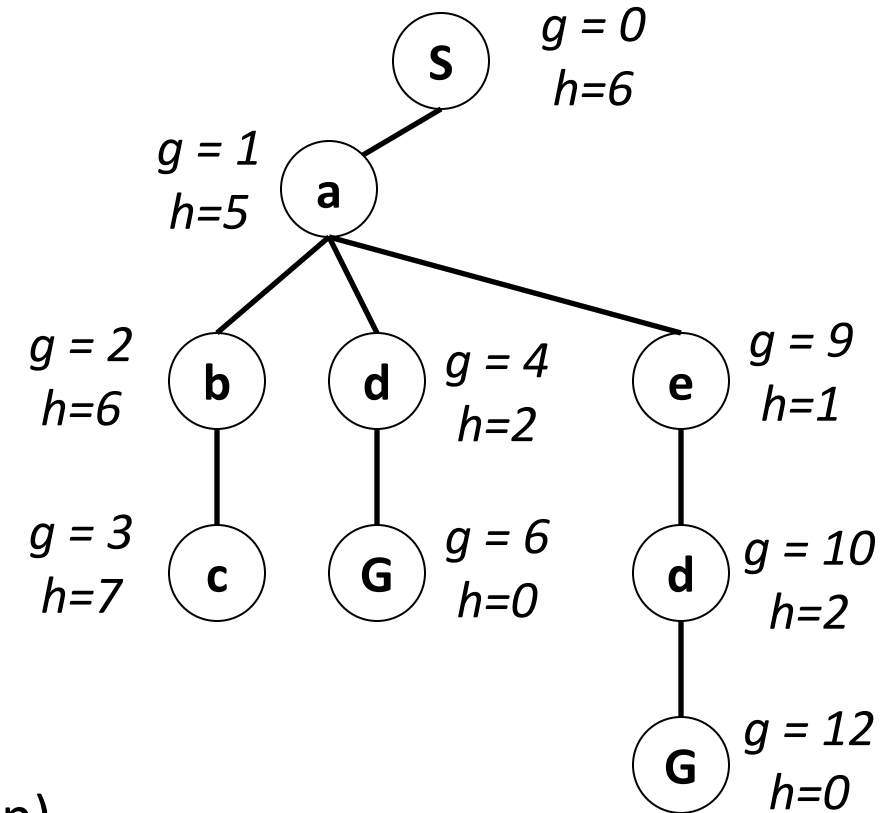
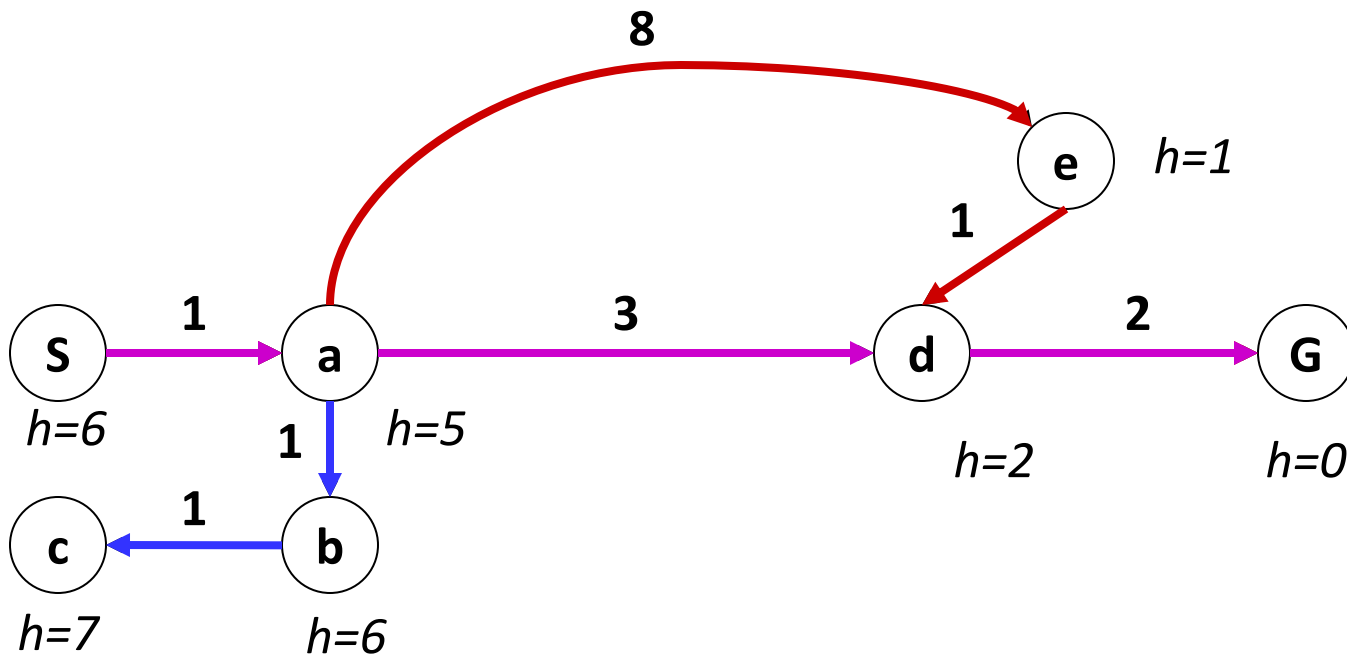
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Combining UCS and Greedy

- **Uniform-cost** orders by path cost, or *backward cost* $g(n)$
- **Greedy** orders by goal proximity, or *forward cost* $h(n)$



- **A* Search** orders by the sum: $f(n) = g(n) + h(n)$

Diskusi

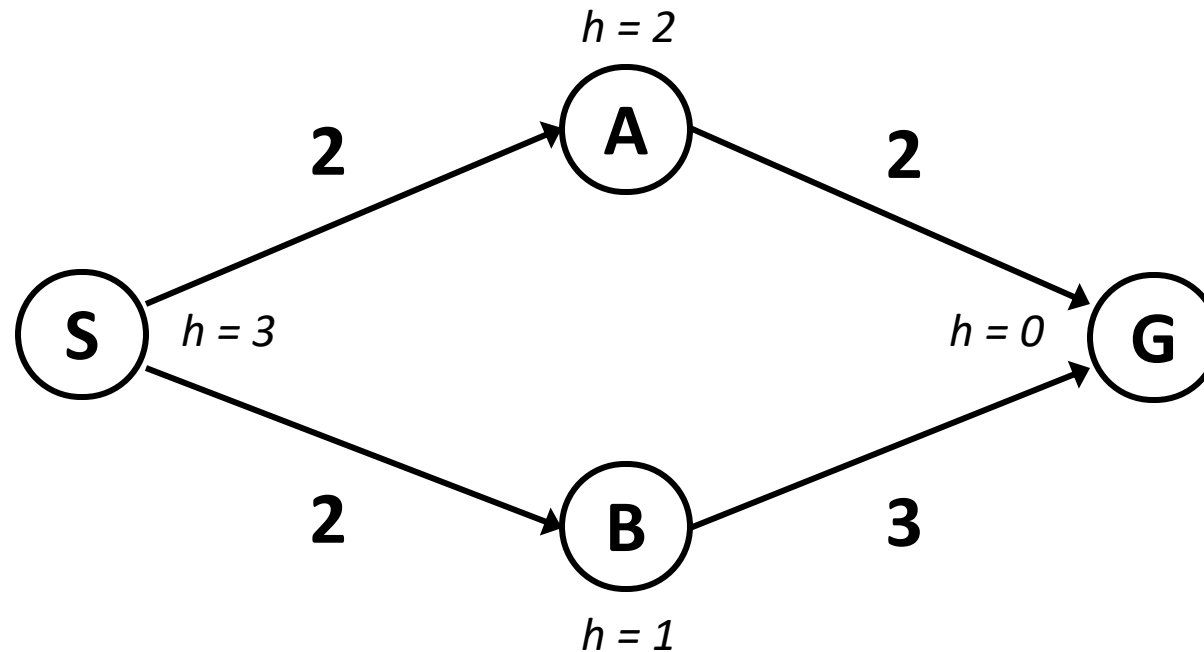
Dalam kondisi seperti apa Algoritme A^ harus berhenti melakukan pencarian?*

Apakah Algoritme A^ optimal?*



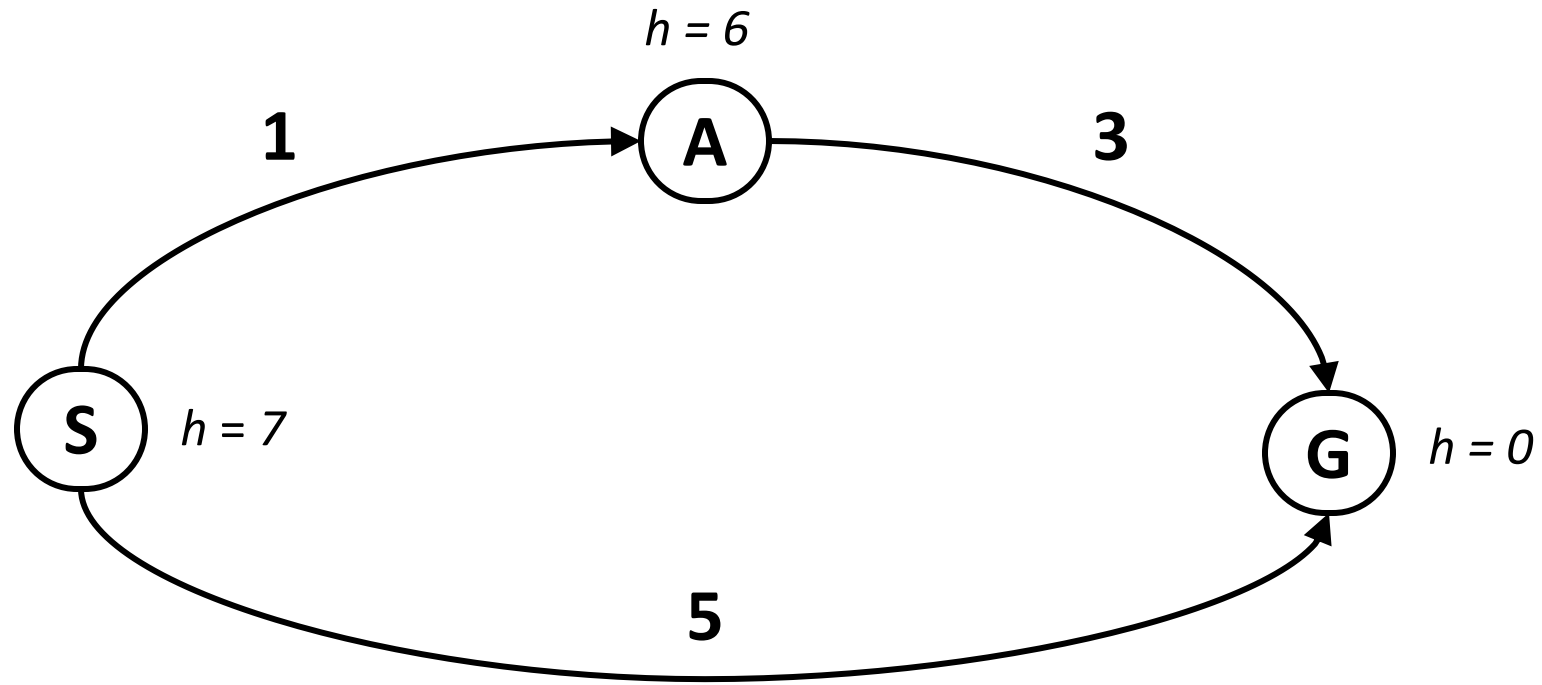
When should A* terminate?

- Should we stop when we enqueue a goal?



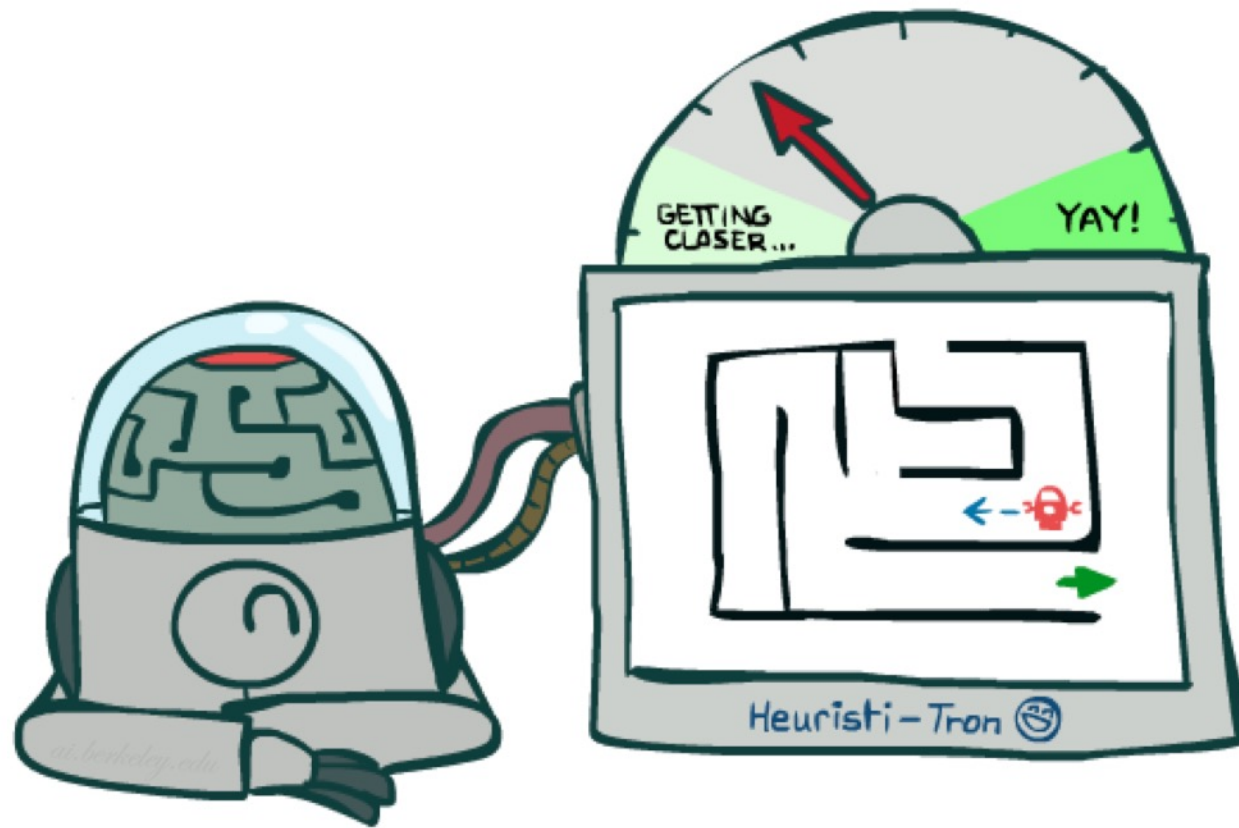
- No: only stop when we dequeue a goal

Is A* Optimal?

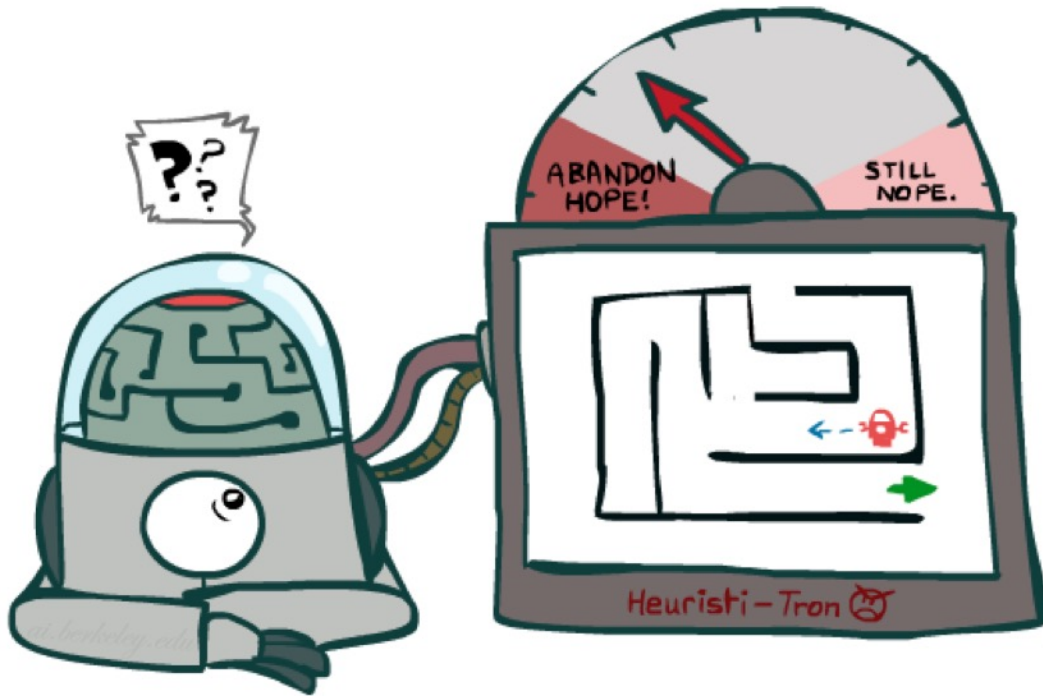


- What went wrong?
- Actual bad goal cost < estimated good goal cost
- We need estimates to be less than actual costs!

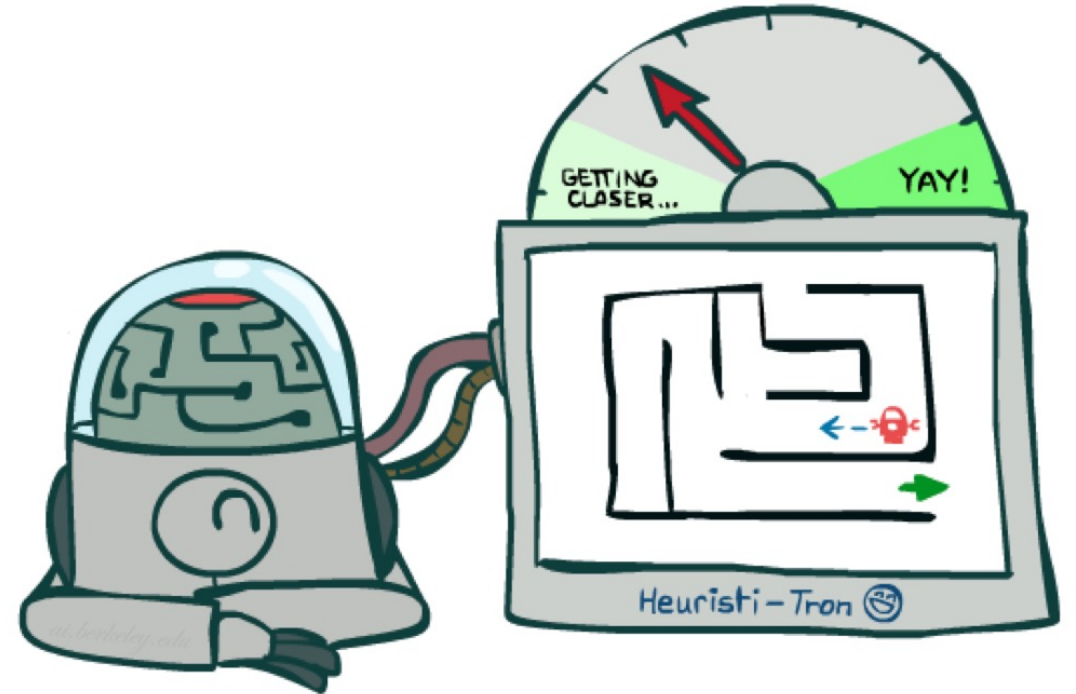
Admissible Heuristics



Idea: Admissibility



Inadmissible (pessimistic) heuristics break optimality by trapping good plans on the fringe



Admissible (optimistic) heuristics slow down bad plans but never outweigh true costs

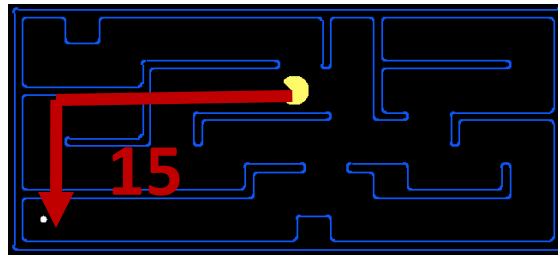
Admissible Heuristics

- A heuristic h is *admissible* (optimistic) if:

$$0 \leq h(n) \leq h^*(n)$$

where $h^*(n)$ is the true cost to a nearest goal

- Examples:



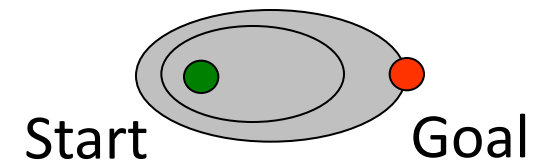
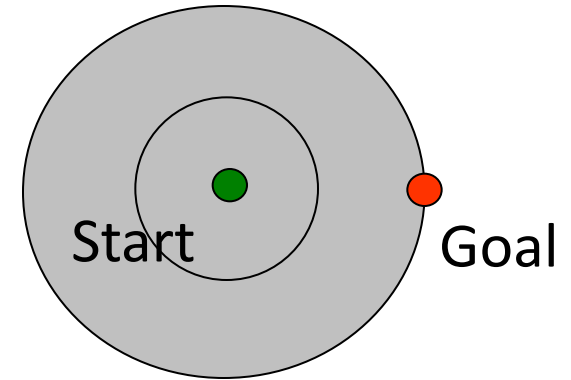
4



- Coming up with admissible heuristics is most of what's involved in using A^* in practice.

UCS vs A* Contours

- Uniform-cost expands equally in all “directions”
- A* expands mainly toward the goal, but does hedge its bets to ensure optimality



[Demo: contours UCS / greedy / A* empty (L3D1)]

[Demo: contours A* pacman small maze (L3D5)]

Video of Demo Contours (Empty) -- UCS



Video of Demo Contours (Empty) -- Greedy



Video of Demo Contours (Empty) – A*



Video of Demo Contours (Pacman Small Maze) – A*



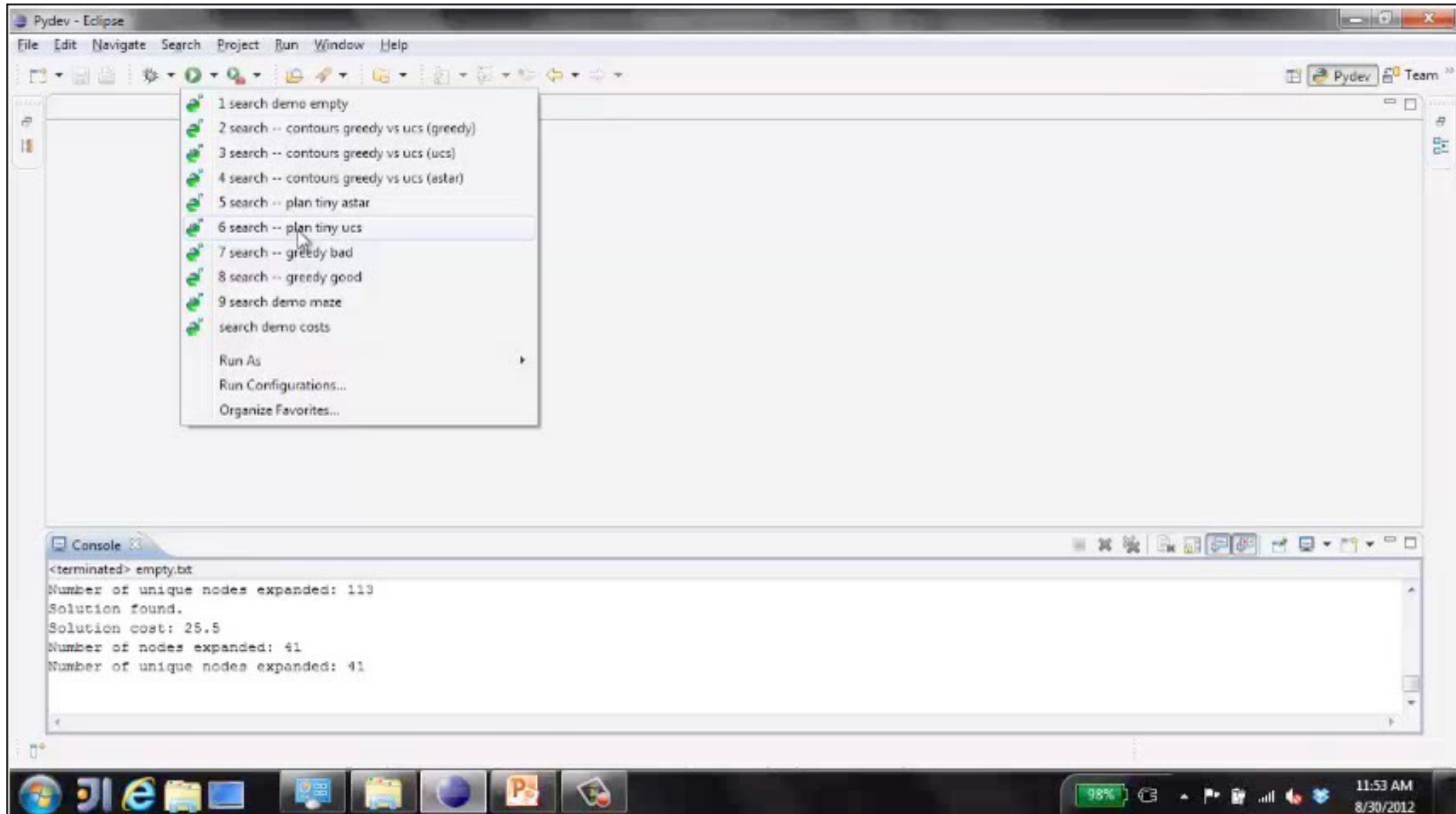
A* Applications

- Video games
- Pathing / routing problems
- Resource planning problems
- Robot motion planning
- Language analysis
- Machine translation
- Speech recognition
- ...

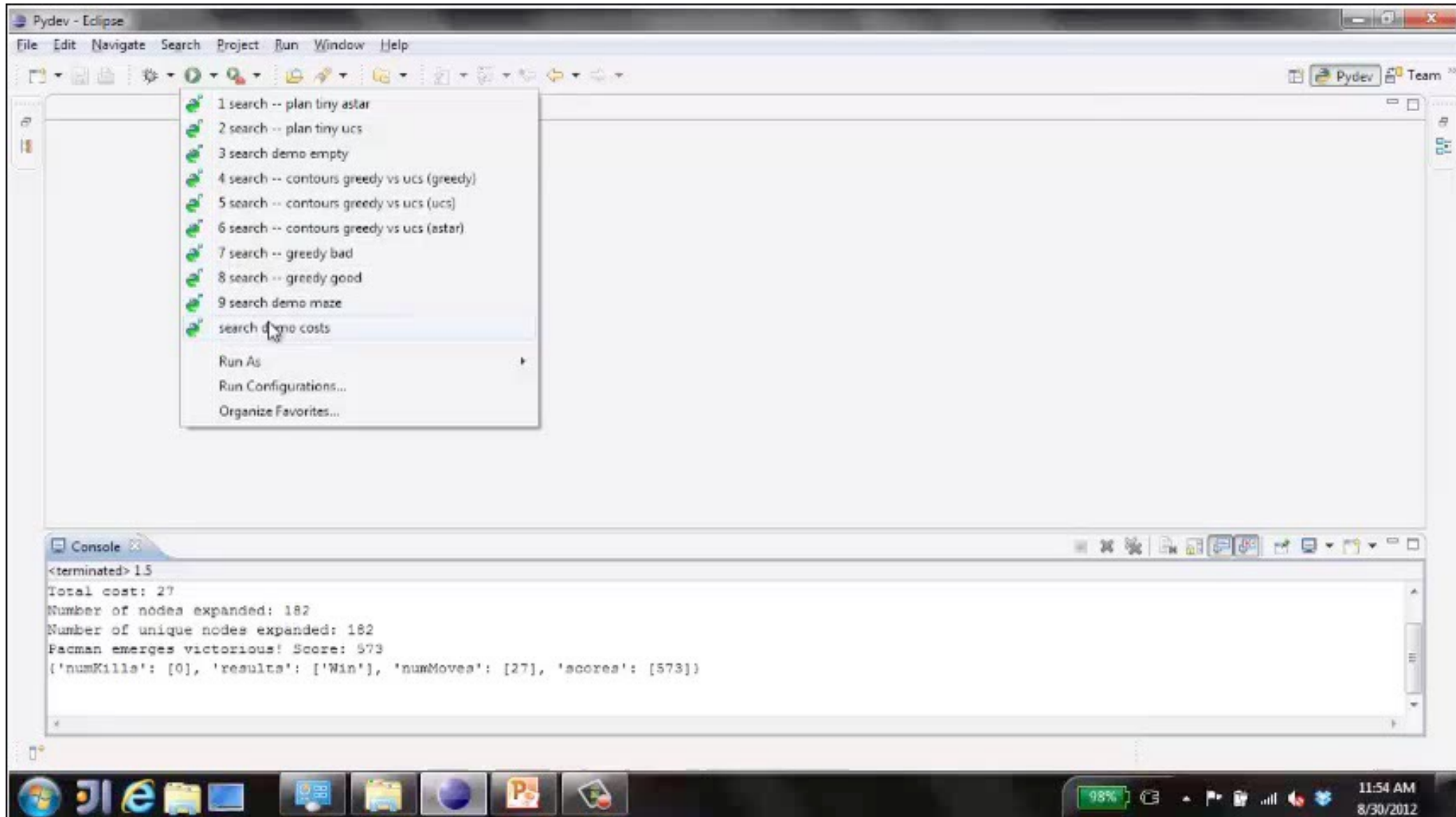


[Demo: UCS / A* pacman tiny maze (L3D6,L3D7)]
[Demo: guess algorithm Empty Shallow/Deep (L3D8)]

Video of Demo Pacman (Tiny Maze) – UCS / A*



Video of Demo Empty Water Shallow/Deep – Guess Algorithm

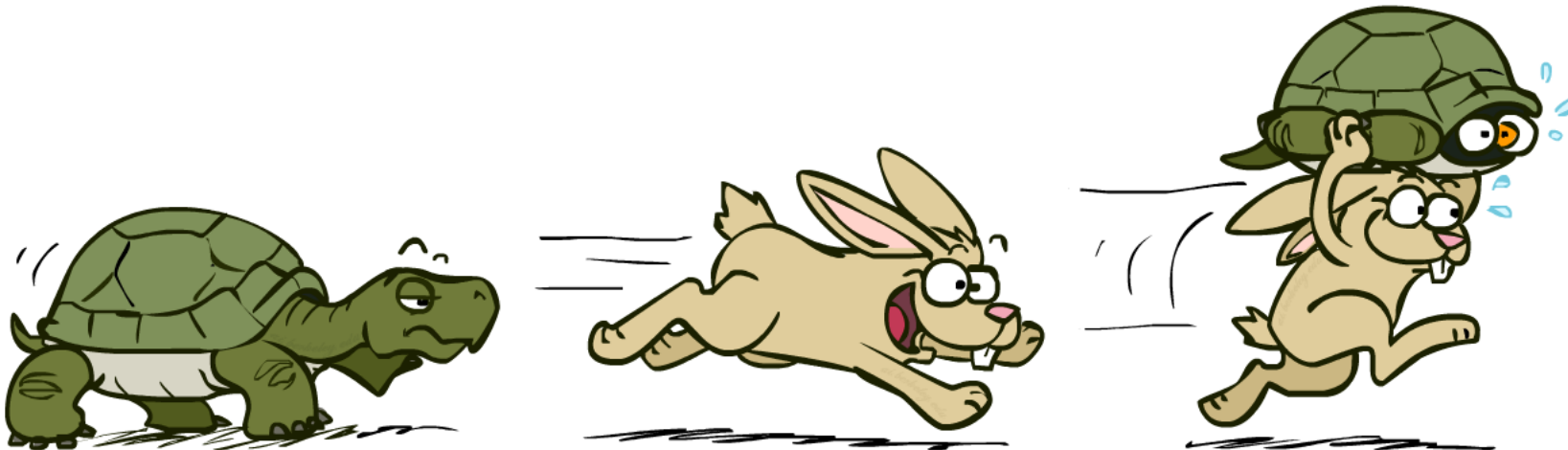


A*: Summary



A*: Summary

- A* uses both backward costs and (estimates of) forward costs
- A* is optimal with admissible / consistent heuristics
- Heuristic design is key: often use relaxed problems



References

- **Russell, S. J., Norvig, P., & Davis, E. (2010). Artificial intelligence: a modern approach. 3rd ed.** Upper Saddle River, NJ: Prentice Hall
- Slides by **Dan Klein** and **Pieter Abbeel** for CS188 Intro to AI at UC Berkeley



Terima Kasih